

Flight School Admin Dashboard — Technical & Functional Audit

SECTION 06-operations-it TYPE audit STATUS **active** OWNER GTM & CS Specialist
UPDATED 2026-08-16

1. Executive Summary & Audit Scope

This document provides a comprehensive audit of the **Flight School Admin & Instructor Dashboard** requirements, bridging the commercial GTM commitments made in `b2b-pipeline.md` and `gtm-schools.md` with the architectural specification in `spec-instructor-dashboard.md`.

Key Audit Findings:

- Commercial Alignment:** The dashboard feature set directly fulfills all promises made on `schools.html` and B2B pilot sales decks (Cohort Exam Readiness %, Cadet Drill-Down, Weak Area Identification, Assigned Reading Paths).
- Architecture & Multi-Tenancy:** The sub-tree Firestore data model (`schools/{schoolId}/roster/{cadetUid}`) effectively preserves 100% per-user isolation of `users/{uid}` logbooks while enabling safe cross-user progress reads.
- Primary Gap Identified:** Study progress currently resides in client-side `localStorage` (`study-progress.js`). Phase A migration to Firestore (`users/{uid}/progress/summary`) is a strict prerequisite before any B2B pilot launch.

2. Multi-Tenant Architecture & Data Isolation Audit

Data Model Architecture

```
schools/{schoolId}
├─ adminUids: string[]
├─ instructorUids: string[]
├─ seatLimit: number
└─ roster/{cadetUid}
    ├─ status: 'invited' | 'active' | 'removed'
    ├─ consent: boolean
    ├─ readiness: number (0-100)
    ├─ lastMockScore: number
    ├─ lessonsDone: number
    └─ assignedPaths: string[]
```

Security Rules Isolation Audit

- **Isolation Principle:** Instructors **never** read into `users/{cadetUid}` . They read only `schools/{schoolId}/roster/{cadetUid}` .
- **Diff Check Enforcement:** Security rules enforce strict field diffing:
 - Cadets can only edit `readiness` , `lastMockScore` , `lessonsDone` , `currencyFlags` , and `progressUpdatedAt` .
 - Cadets can only edit if `roster().consent == true` .
 - Instructors can only edit `assignedPaths` .
 - Seat allocations (`seatLimit` , `seatsUsed`) and roster lifecycle are strictly Cloud Function Admin SDK operations.
- **Audit Assessment: PASS.** The design satisfies security and multi-tenancy rules without exposing user private collections.

3. Data Privacy & KSA PDPL Compliance Audit

Compliance Criteria	Specification Provision	Audit Status	Recommendation
Consent Management	Explicit opt-in modal on first cadet sign-in (<code>consent: true</code>)	PASS	Require re-consent upon annual contract renewal.
Scope Limitation	Only study progress and readiness scores shared; logbooks remain strictly private	PASS	Display explicit privacy badge ("Logbook Private") in cadet view.
Right to Revoke	Cadet can flip <code>consent: false</code> from Settings, immediately blanking shared roster entry	PASS	Ensure Cloud Function clears denormalized progress data on revocation.
DPIA Inclusion	Cross-user data sharing in B2B context	CONDITIONAL	Must update PDPL Data Protection Impact Assessment (DPIA) before pilot launch.

4. Functional Requirements Audit

Core Feature Coverage Matrix

Feature	Pitch / GTM Requirement	Technical Spec Coverage	Readiness
Cohort Readiness Rollup	Display average exam-readiness % across entire cohort	Calculated from <code>roster/{cadetUid}.readiness</code> average	Ready for build

6. Action Plan & Build Roadmap

Phase	Milestone	Priority	Effort	Target Completion
Phase A	Progress Persistence (<code>users/{uid}/progress/summary</code>)	P0 (Blocker)	3 Days	Week 1
Phase B	Firestore Security Rules & Emulator Unit Tests	P0 (Blocker)	4 Days	Week 2
Phase C	Cloud Functions (<code>grantSchoolLicence</code> , <code>createSchool</code> , consent)	P0 (Blocker)	4 Days	Week 2
Phase D	Instructor UI (<code>instructor.html</code> - Cohort View & Drill-Down)	P1	5 Days	Week 3
Phase E	Admin Roster Management & Reading Path Assignment	P1	4 Days	Week 4

Related Documents

- [Instructor Dashboard Design Spec](#)
- [B2B Pipeline Playbook](#)
- [GTM Flight Schools Strategy](#)
- [Customer Health Scoring Spec](#)